Rozwiązania i wskazówki do wybranych przykładów z Z01

$\lim\limits_{n\to\infty}\frac{1}{n^p}=\begin{cases}0 & p>0\\ 1 & p=0\\ \infty & p<0\end{cases}$	$\lim\limits_{n\to\infty}q^n=\begin{cases}\infty & q>1\\ 1 & q=1\\ 0 & \lvert q\rvert<1\end{cases}$
$a>0,\ \lim\limits_{n\to\infty}\sqrt[n]{a}=1$	$\lim\limits_{n\to\infty}\sqrt[n]{n}=1$
$\lim\limits_{a_n\to\infty}\left(1+\frac{A}{a_n}\right)^{a_n}=e^A$	

$\frac{0}{0},\frac{\infty}{\infty},\infty-\infty,0\cdot\infty,\ 1^\infty,0^0,\infty^0$ nieoznaczone

$$\frac{1}{\pm\infty}=0,\quad \frac{1}{0^+}=\infty,\quad \frac{1}{0^-}=-\infty,\quad (0^+)^\infty=0,\quad \frac{\infty}{0^+}=\infty,\ \frac{0}{\infty}=0,\quad \infty^{-\infty}=0$$

0. Dla podanych ciągów napisać wzory określające wskazane wyrazy

$a)\ a_2=\frac{2}{2\cdot 2+3}=\frac{2}{7};\ \ a_{2n+1}=\frac{2n+1}{2(2n+1)+3}=\frac{2n+1}{4n+2+3}=\frac{2n+1}{4n+5}$

$b)\ a_2=\frac{\sqrt{2^2+2}+1}{2\cdot 2}=\frac{\sqrt{6}+1}{2\cdot 2};\ \ a_{2n}=\frac{\sqrt{(2n)^2+2n}+1}{2(2n)}=\frac{\sqrt{4n^2+2n}+1}{4n}$

$c)\ a_3=a_{2+1}=(2+10)!=12!\quad a_m=a_{(m-1)+1}=(m-1+10)!=(m+9)!$

$d)\ a_{k-1}=(k+1)^k;\ a_4=a_{5-1}(5+1)^5=6^5;\ \ a_{1+n}=a_{(n+2)-1}=\big((n+2)+1\big)^{n+2}$
$=(n+3)^{n+2}.$

$g)\ a_{n+1}=\frac{2^{2n+2}}{n+2};\ \ a_n=a_{(n-1)+1}=\frac{2^{2(n-1)+2}}{(n-1)+2}=\frac{2^{2n}}{n+1},\qquad a_{2n}=\frac{2^{4n}}{2n+1}$

$\boldsymbol{h)\ a_{n+2}=\frac{n(n-1)}{n+1};\ \ a_n=a_{(n-2)+2}=\frac{(n-2)(n-3)}{n-1}\quad,\ a_{n-1}=\frac{(n-3)(n-4)}{n-2}.}$

1. Oblicz granice

$b)\ \lim\limits_{n\to\infty}\frac{(\sqrt{n^2+n}+n)}{(\sqrt{n+1}+1)}\ |\ 'n'do\ najwyższej\ potęgi\ z\ mianownika\ ,to\ n^{\frac{1}{2}}\ |$

$$=\lim_{n\to\infty}\frac{\sqrt{n}\left(\sqrt{n+1}+\sqrt{n}\right)}{\sqrt{n}\left(\sqrt{1+\frac{1}{n}}+\frac{1}{\sqrt{n}}\right)}=\infty$$

$c)\ \lim\limits_{n\to\infty}\arcsin\left(\frac{n-1}{n+1}\right)=\arcsin 1=\frac{\pi}{2}$ ponieważ $\lim\limits_{n\to\infty}\frac{n-1}{1+n}=1$

$$d)\ \lim_{n\to\infty}\frac{\operatorname{arctan} n!}{n+1}\left[\frac{\operatorname{arctg}\infty}{\infty}=\frac{\frac{\pi}{2}}{\infty}\right]=0$$

$$e)\ \lim_{k\to\infty}\left(\sqrt{k+1}-\sqrt{k-1}\right)=\lim_{k\to\infty}\frac{\left(\sqrt{k+1}-\sqrt{k-1}\right)\left(\sqrt{k+1}+\sqrt{k-1}\right)}{\left(\sqrt{k+1}+\sqrt{k-1}\right)}=$$

$$=\lim_{k\to\infty}\frac{k+1-(k-1)}{\left(\sqrt{k+1}+\sqrt{k-1}\right)}=\lim_{k\to\infty}\frac{k+1-k+1}{\left(\sqrt{k+1}+\sqrt{k-1}\right)}=\lim_{k\to\infty}\frac{2}{\left(\sqrt{k+1}+\sqrt{k-1}\right)}=0,$$

$$f)\ \lim_{n\to\infty}\frac{3n}{\sqrt{n^2-n}-\sqrt{n^2+n}}=\lim_{n\to\infty}\frac{3n\left(\sqrt{n^2-n}+\sqrt{n^2+n}\right)}{\left(\sqrt{n^2-n}-\sqrt{n^2+n}\right)\left(\sqrt{n^2-n}+\sqrt{n^2+n}\right)}=$$

$$\lim_{n\to\infty}\frac{3n\left(\sqrt{n^2-n}+\sqrt{n^2+n}\right)}{n^2-n-n^2-n}=\lim_{n\to\infty}\frac{3n\left(\sqrt{n^2-n}+\sqrt{n^2+n}\right)}{-2n}=-\infty$$

$$i)\ \lim_{k\to\infty}7^k-6^k-5^k=\lim_{k\to\infty}7^k\left(1-\frac{6^k}{7^k}-\frac{5^k}{7^k}\right)=\lim_{k\to\infty}7^k\left(1-\left(\frac{6}{7}\right)^k-\left(\frac{5}{7}\right)^k\right)=\infty$$

$$j)\ \lim_{n\to\infty}\ln(2n^4-2n+1)-\ln(n^5-2n+1)=\lim_{n\to\infty}\ln\frac{2n^4-2n+1}{n^5-2n+1}$$

$$=\ln\left(\lim_{n\to\infty}\frac{2n^4-2n+1}{n^5-2n+1}\right)\ [\ln 0]=-\infty$$

2. Znajdź granice następujących ciągów $\lim_{n\to\infty}a_n$

$$a)\ \lim_{n\to\infty}\frac{3\cdot 2^{3n-2}-8}{8^{n+1}+16}=\lim_{n\to\infty}\frac{3\cdot 8^n\cdot\frac{1}{4}-8}{8^n\cdot 8+16}=\lim_{n\to\infty}\frac{8^n\left(3\cdot\frac{1}{4}-8\frac{1}{8^n}\right)}{8^n\left(8+16\frac{1}{8^n}\right)}=\frac{3}{32}$$

$$c)\ \lim_{n\to\infty}\frac{(n^4+4)n!+(n-1)!}{n\cdot(n+1)!}=\lim_{n\to\infty}\frac{n!\left[(n^4+4)+\frac{1}{n}\right]}{n!\cdot n\cdot(n+1)}=\infty$$

3. Korzystając z twierdzenia o trzech ciągach oblicz granice $\lim_{n\to\infty}a_n$:

Tw. *o trzech ciągach*
Niech dla $n>n_0$ będzie: $a_n\le b_n\le c_n$ i $\lim_{n\to\infty}a_n=\lim_{n\to\infty}c_n=q$
To: $\lim_{n\to\infty}b_n=q$

$a)\ \lim_{n\to\infty}\frac{\sqrt[n]{4^n+5^n}}{\sqrt[n]{6^n+7^n}}$ licznik i mianownik oddzielnie

1. $\sqrt[n]{5^n}\le\sqrt[n]{4^n+5^n}\le\sqrt[n]{5^n+5^n}=\sqrt[n]{2\cdot 5^n}\to 5$

2. $\sqrt[n]{7^n}\le\sqrt[n]{6^n+7^n}\le\sqrt[n]{7^n+7^n}=\sqrt[n]{2\cdot 7^n}\to 7$

$$\lim_{n\to\infty}\frac{\sqrt[n]{4^n+5^n}}{\sqrt[n]{6^n+7^n}}=\frac{5}{7}$$

b) $\lim\limits_{n\to\infty} \sqrt[n]{5^{n+2}+2^{3n-1}+3^{n+2}} = \lim\limits_{n\to\infty} \sqrt[n]{5^2 5^n + 2^{-1}8^n + 3^2 3^n}$:

prawa nierówność:

$$\sqrt[n]{5^2 5^n + 2^{-1}8^n + 3^2 3^n} \le \sqrt[n]{5^2 8^n + 2^{-1}8^n + 3^2 8^n} \le \sqrt[n]{8^n(5^2+1+3^2)} = 8\sqrt[n]{35} \xrightarrow[n\to\infty]{} 8$$

lewa nierówność:

$$8 \xleftarrow[n\to\infty]{} 8\sqrt[n]{2^{-1}} = \sqrt[n]{2^{-1}8^n} \le \sqrt[n]{5^2 5^n + 2^{-1}8^n + 3^2 3^n}$$

c) $\lim\limits_{k\to\infty} \sqrt[k]{k^2+k}$: $\quad 1 \xleftarrow[k\to\infty]{} \sqrt[k]{k^2} \le \sqrt[k]{k^2+k} \le \sqrt[k]{k^2+k^2} = \sqrt[k]{2}\sqrt[k]{k^2} \xrightarrow[k\to\infty]{} 1$

d) $\lim\limits_{n\to\infty} \sqrt[n]{4^n + 5\cdot 3^{2n} + 2^{n+1}} = \lim\limits_{n\to\infty} \sqrt[n]{4^n + 5\cdot 9^n + 2\cdot 2^n}$

$$9 = \sqrt[n]{9^n} \le \sqrt[n]{4^n + 5\cdot 9^n + 2\cdot 2^n} \le \sqrt[n]{9^n + 5\cdot 9^n + 2\cdot 9^n} = \sqrt[n]{8\cdot 9^n} \to 9$$

e) $\lim\limits_{k\to\infty} \dfrac{k\sin(k!)}{k^2+1}$: $\quad 0 \xleftarrow[k\to\infty]{} \dfrac{k\cdot(-1)}{k^2+1} \le \dfrac{k\sin(k!)}{k^2+1} \le \dfrac{k\cdot 1}{k^2+1} \xrightarrow[k\to\infty]{} 0$

f) $\lim\limits_{n\to\infty} \dfrac{4^n + (-1)^n 2^n}{3\cdot 4^n + 1}$ $\quad \dfrac{1}{3} \xleftarrow[n\to\infty]{} \dfrac{4^n - 2^n}{3\cdot 4^n + 1} \le \dfrac{4^n + (-1)^n 2^n}{3\cdot 4^n + 1} \le \dfrac{4^n + 2^n}{3\cdot 4^n + 1} \xrightarrow[n\to\infty]{} \dfrac{1}{3}$

g) $\lim\limits_{n\to\infty} \dfrac{2n^2 + \sin(n!)}{3n^2 - 4\cos(n!)}$ $\quad \dfrac{2}{3} \xleftarrow[n\to\infty]{} \dfrac{2n^2-1}{3n^2+4} \le \dfrac{2n^2 + \sin(n!)}{3n^2 - 4\cos(n!)} \le \dfrac{2n^2+1}{3n^2-4} \xrightarrow[n\to\infty]{} \dfrac{2}{3}$

h) $\lim\limits_{n\to\infty} \dfrac{5^n + (-2)^n}{3^n + 4}$; $\quad \infty \xleftarrow[n\to\infty]{} \dfrac{5^n - 2^n}{3^n+4} \le \dfrac{5^n + (-2)^n}{3^n+4} \le \dfrac{5^n + 2^n}{3^n+4} \xrightarrow[n\to\infty]{} \infty$

i) $\lim\limits_{n\to\infty} \dfrac{n\sin(n^2) - n^3}{n+1}$; $\quad -\infty \xleftarrow[n\to\infty]{} \dfrac{-n-n^3}{n+1} \le \dfrac{n\sin(n^2)-n^3}{n+1} \le \dfrac{n-n^3}{n+1} \xrightarrow[n\to\infty]{} -\infty$

j) $\lim\limits_{n\to\infty} (n^2 + 2n\cos n)$; $\quad \infty \xleftarrow[n\to\infty]{} n^2 - 2n \le n^2 + 2n\cos n \le n^2 + 2n \xrightarrow[n\to\infty]{} \infty$

k)* $\lim\limits_{n\to\infty} \sqrt[n+1]{3^n + 4^{n+1}}$;

$$4 = \sqrt[n+1]{4^{n+1}} \le \sqrt[n+1]{\frac{1}{3}\cdot 3^{n+1} + 4^{n+1}} \le \sqrt[n+1]{4^{n+1} + 4^{n+1}} \xrightarrow[n\to\infty]{} 4$$

l)* $\lim\limits_{n\to\infty} \sqrt[n+1]{2n+3}$;

$$1 \xleftarrow[n\to\infty]{} \sqrt[n+1]{n+1} \le \sqrt[n+1]{2n+3} = \sqrt[n+1]{2n+2+1} = \sqrt[n+1]{2(n+1)+1} \le \sqrt[n+1]{3(n+1)} \xrightarrow[n\to\infty]{} 1$$

4. Oblicz granice :

$a)$ *metoda I*: $\lim\limits_{n\to\infty}\left(\frac{2n-5}{2n+7}\right)^{5n}=\lim\limits_{n\to\infty}\left[\frac{2n\left(1-\frac{5}{2n}\right)}{2n\left(1+\frac{7}{2n}\right)}\right]^{5n}=\lim\limits_{n\to\infty}\left[\frac{\left(1-\frac{5}{2n}\right)}{\left(1+\frac{7}{2n}\right)}\right]^{5n}=$

$=\frac{\lim\limits_{n\to\infty}\left(1-\frac{5}{2n}\right)^{5n}}{\lim\limits_{n\to\infty}\left(1+\frac{7}{2n}\right)^{5n}}=\frac{e^{\frac{-25}{2}}}{e^{\frac{35}{2}}}=e^{\frac{-25}{2}-\frac{35}{2}}=e^{-30}.$

$b)$ *metoda II*: $\lim\limits_{n\to\infty}\left(\frac{3n+2}{3n-3}\right)^{3n+1}=\lim\limits_{n\to\infty}\left(\frac{3n-3+5}{3n-3}\right)^{3n+1}=\lim\limits_{n\to\infty}(1+\frac{5}{3n-3})^{3n+1}=$

$=\lim\limits_{n\to\infty}\left(1+\frac{5}{3n-3}\right)^{(3n-3)\cdot\frac{3n+1}{3n-3}}=\lim\limits_{n\to\infty}\left(\left(1+\frac{5}{3n-3}\right)^{3n-3}\right)^{\frac{3n+1}{3n-3}}=(e^5)^1=e^5.$

ponieważ : $\lim\limits_{n\to\infty}\frac{3n+1}{3n-3}=1$

$c)$ $\lim\limits_{n\to\infty}\left(\frac{3n^2+2}{3n^2-3}\right)^{3n^2+2n-3}=\lim\limits_{n\to\infty}\left(\frac{3n^2-3+5}{3n^2-3}\right)^{3n^2+2n-3}=\lim\limits_{n\to\infty}\left(1+\frac{5}{3n^2-3}\right)^{3n^2+2n-3}=$

$\lim\limits_{n\to\infty}\left[\left(1+\frac{5}{3n^2-3}\right)^{3n^2-3}\right]^{\frac{3n^2+2n-3}{3n^2-3}}=e^5$ bo $\lim\limits_{n\to\infty}\frac{3n^2+2n-3}{3n^2-3}=1$

$e)$* $\lim\limits_{n\to\infty}n(\ln(n+1)-\ln n)=\lim\limits_{n\to\infty}n\ln\frac{n+1}{n}=\lim\limits_{n\to\infty}n\ln(1+\frac{1}{n})=\lim\limits_{n\to\infty}\ln\left(1+\frac{1}{n}\right)^n=\ln e=1$

$f)^*$ $\lim\limits_{n\to\infty}\left(1+\frac{1}{n}+\frac{1}{n^2}\right)^{3n}=\lim\limits_{n\to\infty}\left[\left(1+\frac{n^2+n}{n^3}\right)^{\frac{n^3}{n^2+n}}\right]^{3n\cdot\frac{n^2+n}{n^3}}=e^3$

$g)^*$ $\lim\limits_{n\to\infty}\left(0{,}99+\frac{1}{n}\right)^n\ [0{,}99^\infty]=0$

ponieważ: $\lim\limits_{n\to\infty}\left(0.99+\frac{1}{n}\right)=0{,}99$ oraz $\lim\limits_{n\to\infty}(0.99)^n=0.$